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What is the body without organs? Machine and organism in Deleuze and Guattari

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Abstract In the two volumes which make up *Capitalism and Schizophrenia*, Deleuze and Guattari propose new concepts of "machine" and "organism." The problem of the relationship between machines and organisms has a long philosophical history, and this essay treats their work as a contribution to this debate. It is argued that their solution to this problem is found in their difficult concept of the "body without organs," a concept that is given some much-needed clarification in the essay. The first section details Deleuze and Guattari's concept of the machine, examining the ways in which it differs from the traditional concept as described by Canguilhem: (1) their machines do not have predictable movements, but instead produce events; (2) they do not have a purpose; (3) they are able to reproduce themselves. The second section details their conception of the organism through their account of Geoffroy Saint-Hilaire: (1) organisms are bodies which normalize and which create hierarchies; (2) they also do not have a purpose; (3) they have a "unity of composition." The final section argues that their concept of the "body without organs" shows us how to understand the relation between the two transformed concepts, and defines the body without organs as the becoming-machine of the organism.

Keywords Body without organs · Machine · Organism · Deleuze · Event · Virtual · Canguilhem · Mechanism · Vitalism · Machinism · Inorganic vitalism

In the two volumes which make up *Capitalism and Schizophrenia*, Deleuze and Guattari propose new concepts of "machine" and "organism." The problem of the relationship between machines and organisms has a long philosophical history, and my aim in this essay is to treat this part of their work as a contribution to this debate. Deleuze and Guattari's intervention re-organizes the cluster of concepts typically

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associated with each term, bringing certain features usually thought to be exclusively "organic" into their concept of "machine," and of bringing certain features usually thought to be "machine-like" into their concept of "organism." Their re-working of these concepts leads them to a very different understanding of their relation: their position is neither a classical mechanism nor a classical vitalism, although it has affinities with both. With the vitalist, for example, they share a distrust of mechanism's understanding of reality as predictable and measurable. With the mechanist, they share a distrust of vitalism's idea of a special kind of matter particular to the organic. If classical vitalism contrasted the "good" creative, spontaneous organism to the "bad" inert, lifeless machine, Deleuze and Guattari will emphasize, on the contrary, the ability of the machine to create something new, and also the normalizing and constraining aspects of the organism. Where some vitalist or neo-vitalist philosophers reverse the traditional order of precedence, re-inscribing "life" as the dominant term over and against the "machine," Deleuze and Guattari go a step further, advancing their own concept of the "body without organs" as a new way of understanding the relation between the two.[1]

The first part of this essay deals with the concept of "machine." By examining three key features of their conception of the machine, I show how Deleuze and Guattari's account differs from classical mechanism. First, I examine their account of the unpredictability of machines, specifically the way they understand them to produce "events." Second, I show how their machines are non-teleological in the sense that they are not structured according to a governing function or purpose. Third, through their discussion of Samuel Butler, I show how they understand machines to have reproductive systems. The second part of the essay deals with the concept of "organism." First, I explain the basic feature of their account of organisms, namely that the term describes not some specifically vital kind of "matter," but rather describes bodies which are organized in a particular way. Second, through their dramatization of the debate between Geoffroy Saint-Hilaire and Georges Cuvier, I investigate the way they re-conceptualize the relation between form and function in organisms. Third, I show how they take up Geoffroy's project of "transcendental anatomy," which posits a fixed plan or schema which is shared across different organisms. The third and final part of the essay deals with their criticisms of the organism, and explains their notorious idea of the "body without organs." This concept, I argue, is their account of the relation between machines and organisms, and can be defined as the becoming-machine of the organism.

Before turning to the specifics of Deleuze and Guattari's account, I would first like to give a fuller account of the problem to which it is a response, namely the problem of the relationship between machines and organisms. To do this, I will use a lecture by Canguilhem, one of Deleuze's early teachers, entitled "Machine and Organism."[2] This lecture gives a thorough account of the philosophical history of the problem, and of the classical view that Deleuze and Guattari pit themselves

[1] There is a lot of terminology developed within *Capitalism and Schizophrenia*, which can make it difficult to understand the ideas without making constant reference to the whole web of other concepts. In this essay, I try to keep the Deleuzo-Guattarian technical vocabulary to a minimum, sticking as closely as possible to the problem at hand. As such, I will leave the more intricate terminological issues to footnotes.

[2] Canguilhem (2009). For a very helpful reading of Canguilhem's lecture, see also Hacking (1998).

against, and so will serve as a useful entry-point into the sections that follow. Canguilhem shows that the two concepts, machine and organism, have typically been seen as relating to two different domains, and have been associated with two contrasting series of features. The first feature is the movement of the body in question: Canguilhem contrasts the "geometric and measurable" movements of machines, which can always be predicted or calculated in advance, with the unpredictable spontaneity of the organism.[3] The second feature relates to the question of teleology. Although mechanism is often praised for eliminating the need for teleological explanations in biology, Canguilhem notes that there is nonetheless a certain residual teleology still tied to the metaphor of the machine. Machines are always designed to "achieve specific ends," and thus retain a certain function or purpose latent within their very structure, as if "implanted" by their creator or designer.[4] Although there have been many different answers to the question of organic teleology, the standard view, at least in the post-Darwinian world, is that organisms should be explained without reference to any intrinsic inner "purpose." The final feature concerns reproduction. Whereas organisms are able to reproduce themselves, "machines do not construct other machines," depending instead on something external and non-mechanical for their creation and continuing reproduction.[5]

Canguilhem begins his lecture by noting that "the relationship between machine and organism has generally been studied in only one way. Nearly always, the organism has been explained on the basis of a preconceived idea of the structure and functioning of the machine."[6] As he suggests, the fundamental question that has generally been at stake in the debate between mechanism and vitalism is whether or not those phenomena which seem to be specifically "vital" can nonetheless be explained in exclusively mechanistic terms. Is it possible to "explain away" the differences just mentioned, thus collapsing the distinction between machine and organism? The classical response to this question is Descartes' theory of the animal-machine, which takes the animal to be without a soul and hence qualitatively no different from other mechanical phenomena, like the "clocks, artificial fountains, and other such machines" that so fascinated him and many others in his era.[7] Every organ was to be re-described by analogy with some kind of technical device, an undertaking carried out across the eighteenth-century life sciences, which spawned a host of such analogies: lungs described as bellows, muscles as a complex system of pulleys, teeth as pincers, arteries as hydraulic tubes and so on.[8]

Of course, there have been many criticisms of Descartes' theory, coming from many different quarters. But the issue I would like to focus on, which will serve as the point of entry into our discussion of Deleuze and Guattari, is that Descartes never actually defines what he means by "machine," even though it is

[3] Canguilhem (2009, p. 46).

[4] Canguilhem (2009, p. 54).

[5] Canguilhem (2009, p. 55).

[6] Canguilhem (2009, p. 45).

[7] See Descartes (1998, p. 99).

[8] Canguilhem (2009, p. 47). See also Gasking (1967, pp. 167–168).

the central metaphor of his philosophy. As Des Chene puts it in a useful study, Descartes simply "takes the intelligibility of machines for granted," and explains organic phenomena through this analogy.[9] What this means, then, is that certain un- or under-thematized features of machines secretly govern the way the metaphor functions in Descartes, in ways which ended up causing him problems which he was never able to definitively overcome. One such feature, mentioned above, is machines' inability to reproduce. If one explains organisms by analogies with machines, but the conception of "machine" one is working with is one which makes it impossible to understand how it could reproduce, then one will be unable to account for reproduction in organisms. And indeed, despite working on it extensively, Descartes was never able to give a satisfactory account of the generation of animal bodies.[10] The problem of how to explain generation continued to plague mechanistic theories of the organism well into the nineteenth century.[11]

I suggest, then, that any investigation into the relation between these two things, machines and organisms, ought to begin by reflecting on the meaning of each term. Canguilhem's historical study of these two concepts leads him to propose the interesting idea of reversing the traditional question: he asks not whether one can account for organisms in terms of the machine, but whether one can account for machines "as if they were extensions of human behavior or life processes," and hence "inscribe the mechanical into the organic."[12] Whilst there is much of interest in this proposal, Deleuze and Guattari lead us in a different direction. Rather than simply reversing the order of precedence, their strategy is to re-conceive the original terms, and then to propose a new concept to account for this transformed relation; as we will see, this leads them to a valorization of the machine, a criticism of the organism, and the development of their own concept of the "body without organs," which, I argue, can be defined as the becoming-machine of the organism.

1 What is a machine?

In a useful essay, François Dosse argues that the concept of "machine" came more from Guattari than from Deleuze. Citing Guattari's early essay "Machine and Structure," he explains that the original intention behind the concept was strategic:

[9] Des Chene (2001, p. xi, see also p. 68). Canguilhem makes much the same point throughout his lecture.

[10] See chapter 2 of Des Chene (2001, pp. 35–52), which is devoted to this problem. See also the useful summary in Gasking (1967, pp. 67–69).

[11] Malebranche, for example, thought that Cartesian principles led one to a very different account of generation than the one proposed by Descartes himself. His Cartesianism led him to adopt a particularly extreme version of preformationism according to which each embryo already contains its whole line of descent pre-formed within itself, like a Russian doll: "the body of every man and beast born till the end of time was perhaps produced at the creation of the world" (Malebranche 1997, p. 27). This solution avoids the problem of generation even more ingeniously than Descartes', in that it removes animal generation from the physical sphere entirely, explaining it instead as a pre-given and inscrutable act of God.

[12] Canguilhem (2009, pp. 63–64).

"'Machine' is a veritable catchphrase that was meant to dethrone another catchphrase of the time: the notion of 'structure.'"[13] In contrast to the idea of "structure," which seems to suggest something fixed or static, activity seems to be built into the very notion of a "machine." Guattari's focus is therefore not on the aspect of machines typically stressed by mechanists, that is, the predictability of their movements, but rather is on the fact that they are *productive*, that they "do" something. Dosse explains further: "against structure, which is defined by its ability to exchange its particular elements, the machine would stress repetition, but in the sense in which Deleuze understands it—that is, repetition as difference."[14] One of the great insights of Deleuze's *Difference and Repetition* was that there exist forms of repetition which do not only produce a series of identical terms; the kind of repetitions that Deleuze is interested in are those which introduce *difference* into the thing they repeat. Consider, for example, the various performances of a piece of music, each of which constitutes a "repetition" of the notation written out by the composer. In each performance, the same thing is repeated (the written score), but the work that is produced can vary enormously, whether because it is played by different musicians, conceived by a different conductor or musical director, or even played differently by the same musicians on different nights.[15] Guattari's idea was to re-conceive the notion of "machine" in line with this Deleuzian notion of repetition, to arrive at a concept which would explain the production of differences. However, this example comes from the artistic, human sphere, where the production of difference is "built in" to repetition, which we would usually contrast with the mechanical sphere, where what is repeated in each new production would seem to be the same each time. How is it, then, that machines can contribute to the production of something genuinely *different*? How can we get from a calculating machine to something of the order of the "event?"[16]

The answer to these questions lie with certain basic features of Deleuze's ontology. It is certainly the case that most machines are produced with a certain purpose in mind, and, when put to work, will carry out that purpose in a predictable manner. There is an aspect of dead repetition of the same within the productive activity of the machine. But what is not predictable in advance are the *capacities* that a machine has, which only emerge once it enters into combination with other machines. One can enumerate all of the material features of a certain machine, one can perfectly describe how it works in mechanical terms, but this will never result in an exhaustive account of all the possible ways its operations would be changed if it were put into different contexts, or combined with other machines. Capacities do not only depend on the form or structure of the machine—as Deleuze and Guattari ask at the beginning of *Anti-Oedipus*: "can we possibly guess, for

[13] Dosse (2012, p. 135), citing Guattari (1984) .

[14] Dosse (2012, p. 127).

[15] The example Deleuze gives at the opening of *Difference and Repetition* is Monet's series of "Water Lilies" paintings; in each case, the subject is the same, such that the paintings constitute repetitions of one another, but these are repetitions which "make a difference" in the sense that the work produced is different each time (Deleuze 1994/1968, p. 1/8). When citing texts by Deleuze, or by Deleuze and Guattari, I will cite first the English pagination, then the French.

[16] A term Guattari uses many times in his essay.

instance, what a knife rest is used for if all we are given is a geometrical description of it?"[17]—but also on the capacities of the other machines it could potentially enter into relation with. As such, one cannot predict with certainty how it will react before it is actually put into those relations. A knife-rest is obviously built with a certain purpose in mind, but there is nothing in its structure which prevents it from being used in different contexts, to carry out different functions. One could perfectly understand what the knife-rest is in terms of its actual properties, but this does not at all exhaust what it can *do*, and for Deleuze and Guattari, following Spinoza, "we know nothing about a body until we know what it can do [*ce qu'il peut*], in other words, what its affects are, how they can or cannot enter into composition with other affects, with the affects of another body."[18]

Thus, whereas Cartesian mechanism has tended to emphasize the formal and substantial aspects of machines, the actual properties they have, and the predictable movements that this entails, Deleuze and Guattari's "machinism" emphasizes instead the "virtual" side, the capacities that machines have to do something *other* than what they were designed to do. They do not understand machines to have been built "for" a specific end, which would thus remain within them as a kind of latent purpose or purposiveness; for them, a machine does not have a "purpose" prior to its being actually used to carry one out. Their machines always have a multiplicity of functions, which do not depend on the intentions of their creator but on their relations with other machines, and hence cannot be exhaustively enumerated in advance. Like classical mechanists, they use the machine as a paradigm to understand all of reality (they call their position a "universal machinism")[19], but as we can already see, the import of this comparison is quite different. One of the common worries that is raised about mechanistic philosophy is that it "explains away" seemingly non-mechanical ideas like chance, freedom or novelty, but what Deleuze and Guattari are interested in with the concept of "machine" is precisely the capacity it has *not* to be mechanical, that is, the way in which it can produce something new and unexpected. Rather than opposing the creative flow of life to the lifeless predictability of the machine, Deleuze and Guattari develop a theory of the machine connected to the "event."

As I have suggested, the notion of "event" in Deleuze is his way of understanding the creation of something new. However, this is only one side of his account. As well as moving from conditions to the conditioned such that, for example, a combination of two machines can lead to the production of something which has never been produced before, Deleuze's event also works in the other direction, such that the event modifies its own conditions, transforming those very things which made it happen in the first place. An illustrative example of this difficult idea can be found in Borges' essay "Kafka's Precursors." In this essay, Borges studies Kafka's "precursors," that is, those writers who seem to anticipate or prefigure aspects of his unique style. After enumerating a number of different writers, from Zeno through Han Yu to Kierkegaard, Borges says the following:

[17] Deleuze and Guattari (2000/1973, p. 3/8).

[18] Deleuze and Guattari (1987/1980, p. 257/314).

[19] Deleuze and Guattari (1987/1980, p. 256/313).

> If I am not mistaken, the heterogeneous pieces I have enumerated resemble Kafka; if I am not mistaken, not all of them resemble each other. This second fact is more significant. In each of these texts we find Kafka's idiosyncrasy to a greater or lesser degree, but if Kafka had never written a line, we would not perceive this quality; in other words, it would not exist.[20]

According to Borges, it is not only that something new and unique emerged when Kafka brought together this unique series of precursors in the production of his major works. Also, and perhaps more interestingly, Kafka's writing *creates* new connections between these earlier writers, connections which would have remained utterly imperceptible and hence non-existent had Kafka never written his own work. The event of Kafka's work affects how we understand these earlier writers' places in the history of literature, and changes how we understand and read their work in turn, bringing out new aspects of their style which would not have been visible were it not for Kafka. The Kafka-event is thus *both* the singularity of his work as a mixing together of his unique combination of influences, but is *also* something that works in the other direction, modifying in turn those very influences. It is in this sense that Borges writes that: "every writer creates his own precursors. His work modifies our conception of the past, as it will modify the future."[21] Thus, when a machine or combination of machines produces an event, it does not simply remain what it was before, but is itself transformed in the process.[22]

The third traditional feature of machines that Deleuze and Guattari undermine is their purported inability to reproduce. This part of their account is developed in a discussion of Samuel Butler's remarkable text "The Book of the Machines."[23] We have already seen how the inability of machines to reproduce themselves caused problems for Descartes' theory: if machines cannot reproduce, and one uses machines as one's basic principle of explanation, then one will be unable to explain reproduction. Butler's argument, however, is that machines *do* reproduce, and the

[20] Borges (1964, p. 191).

[21] Borges (1964, p. 192).

[22] The notion of "event" is a crucial one not only in Deleuze, but also in other philosophers of his generation. These others also not only create a new concept of "event," but also and perhaps even more interestingly, follow Deleuze and Guattari in connecting it to the concept of "machine." In Derrida, for example, there is an attempt to think the apparently impossible thought of the "event-machine," that is, a concept that would bring together these two apparently "antinomic" orders: "Will this be possible for us? Will we one day be able, and in a single gesture, to join the thinking of the event to the thinking of the machine? Will we be able to think, what is called thinking, at one and the same time, *both* what is happening (we call that an event) *and* the calculable programming of an automatic repetition (we call that a machine)?" (Derrida 2002, p. 72). For an examination of Derrida's version of the concept of "event", see my "An Event Worthy of the Name, a Name Worthy of the Event" (Smith 2015).

[23] Deleuze and Guattari (2000/1973, pp. 283–286/337–341). The following quotations are from Butler, "The Book of the Machines," which is a section of his 1872 novel *Erewhon*. In these chapters, the protagonist is speaking to a mad professor, who is convinced that machines are "evolving" at an unprecedented rate (the book is, in part, a parody of Darwinism), and advocates the destruction of all machines to prevent what he sees as their inevitable future domination of mankind. Butler has the professor respond to criticisms of this theory, criticisms which mostly revolve around the "organic" capacities he seems to be illegitimately attributing to machines, one of which is their capacity for reproduction.

first step in his argument is to claim that we only don't see this because we are expecting to see a resemblance between their reproductive systems and our own:

> It is said by some with whom I have conversed upon this subject that the machines can never be developed into animate or quasi-animate existences inasmuch as they have no reproductive system, nor seem ever likely to possess one. If this be taken to mean that they cannot marry, and that we are never likely to see a fertile union between two vapor-engines with the young ones playing about the door of the shed, however greatly we might desire to do so, I will readily grant it. But the objection is not a very profound one. No one expects that all the features of the now existing organizations will be absolutely repeated in an entirely new class of life.[24]

Butler goes on to argue that a closer analogy for the reproduction of machines might be found in ants or bees: the vast majority of ants or bees are not directly part of the reproductive process, but this does not lead anyone to the conclusion that ants or bees do not have the capacity for reproduction. The work that these non-reproductive individuals do is crucial to the reproduction of the community as a whole (the queen bee would not be able to reproduce were it not for the work of the worker bees, for example), and so, Butler argues, we should simply widen our concept of reproduction such that it encompasses the entire community as a reproductive system. The only reasonable criterion for deciding whether or not machines have reproductive systems, then, is whether they can systematically reproduce themselves, and as Butler asks: "how few of the machines are there which have not been produced systematically by other machines?"[25] Of course, this reproduction was aided by humans, but, to continue the analogy: "is it not insects that make many of the plants reproductive?" Butler argues that, for many organisms, reproduction is "effected by a class of agents utterly foreign to themselves"; his example is the red clover, which is only able to reproduce with the aid of the bumble bee.[26] Since we would want to say that the red clover does reproduce, we would have to include the bumble bee as a part of that expanded reproductive system. Why not argue, then, that we humans are simply a part of the reproductive system of machines?[27]

[24] Butler (1986, p. 188).

[25] Butler (1986, p. 188).

[26] Butler's example is thus close to Deleuze and Guattari's often repeated example of the wasp and the orchid.

[27] Butler, in a passage cited in *Anti-Oedipus*, makes the further point that the offspring of machines do not have to resemble their "parents": "The truth is that each part of every vapor-engine is bred by its own special breeders, whose function is to breed that part, and that only" (Butler 1986, p. 190). Just because thimbles do not appear to directly produce other thimbles (to use another of Butler's examples), does not mean that they do not have a reproductive system. In Deleuze and Guattari's terminology, we could say that their reproduction is effectuated by the *abstract machine of the thimble*, which comprises everything that it takes for the thimble to be able to continue to reproduce itself. This abstract machine includes the concrete machines that work towards their production, the factory in which they are produced, the humans engaged in their use, sale and maintenance, but also all the related practices, discourses and histories: the aesthetics of the thimble, the political economy of sewing.

2 What is an organism?

For Deleuze and Guattari, organisms are not defined according to certain "organic" features or properties that are unique to them as a peculiar kind of substance: "there is no vital matter specific to the organic."[28] The word organism, rather, describes a type of body that is organized in a certain way, namely one that is "centralized," "hierarchized," and "self-directed."[29] The "organs," as we have already seen, are understood by Deleuze and Guattari on the model of the machine, and the organism is the higher-order construction which holds the organs together, giving them a unified, regularized form (they twice use the phrase "the organization of organs we call the organism").[30] The organism, then, is not some "vital impulse," but a process which holds together the otherwise disjointed, scattered collection of organs/machines. Of course, other higher-order machines are also able to impose forms on the lower-order machines out of which they are constituted, but "organism" names a specific kind of organization, one which is self-regulating.

Deleuze and Guattari develop an important part of their theory of the organism through the mouthpiece of Geoffroy Saint-Hilaire, a nineteenth-century biologist who took up the project of "transcendental anatomy."[31] Deleuze and Guattari provide a fantastical dramatization of the acrimonious public debate that he had with Cuvier, and it is in this context that they ask us to "sing the praises" of Geoffroy Saint-Hilaire. According to Cuvier and his school of anatomy, organisms form a single, unified system. Each of the parts is in reciprocal relation with all the other parts, and all are governed by certain pre-existing laws which ensure that they are composed in a harmonious, well-ordered unity. This presumption of a pre-existing harmony of the organism as a whole led Cuvier to the hypothesis that: "the nature and distribution of the parts of the organism can be discovered simply by reasoning from their place in the functional unity of the organism."[32] Following on from this hypothesis, Cuvier boasted that, given even just a small bone, he would be able to reconstruct the anatomy of the entire animal. This meant, then, that in his school of "comparative anatomy," one would compare organs across species using the principle of *analogy*, which makes comparisons on the basis of shared functions (for example, the legs of a crab are analogous to the legs of a horse).[33]

[28] Deleuze and Guattari (1987/1980, p. 45/61).

[29] As described in Protevi (2005, p. 200). See also Protevi (2012).

[30] Deleuze and Guattari (1987/1980, pp. 158/196; 163/201).

[31] See Deleuze and Guattari (1987/1980, pp. 45–47/60–63). For a good account of the debate between Cuvier and Geoffroy, see Appel (1987). For a very helpful account of how this debate bears on Deleuze's thought, see chapter 8 of Somers-Hall (2012).

[32] Somers-Hall (2012, p. 222).

[33] Somers-Hall notes that Hegel relied extensively on Cuvier in his philosophy of nature. See e.g. the following quotation from Cuvier, cited in Hegel's *Philosophy of Nature* §370 addition: "Every organized creature forms a whole, a *unified* and closed system, all the parts of which mutually correspond, and by reciprocal interaction upon one another contribute to the same purposive activity. None of these parts can alter without the others altering too; and consequently each of them, taken on its own, suggests and gives all the others" (cited in Somers-Hall 2012, p. 222). We therefore might want to read Deleuze and Guattari's staging of the debate between Cuvier and Geoffroy as a coded critique of Hegel. Somers-Hall sees their respective theories of the organism as the key site of contestation between the systems of Hegel

Geoffroy, by contrast, did not assume any pre-existing harmony of the organism, and rejected all explanations based on function. Rather than seeing *analogies* between the organs of different species, he saw *homologies*, which compare them in morphological terms, or more specifically how they came to have the forms that they do.[34] An analogy is "a part or organ in one animal which has the same function as another part or organ in a different animal," whereas a homology is "the same organ in different animals under every variety of form and function."[35] The leg of the crab is thus *not* "homologous" to that of the horse, because it was formed in a completely different way, according to a completely different set of processes. A human arm and a bat's wing, however, *are* homologous, even though their functions could not be more different (see Fig. 1).

We can now see, following on from Deleuze and Guattari's concept of the machine, why they would prefer the principle of homology to that of analogy: an analogy between different organs presupposes that each organ has one single purpose, given in advance; a homology, by contrast, does not presuppose any pre-existing purpose, and allows organs to have an unlimited variety of different functions.

The second important feature of Geoffroy's account (its "transcendental" aspect) is that, even as he emphasizes the great diversity of functions of these organs, he sees these different forms as nonetheless coming from one single schema, which is actualized in a multiplicity of different ways. The homologies on display in Fig. 1 come about, according to Geoffroy, because the schema out of which these multiple actual entities emerged is *the same* in each case: "all vertebrates [are] formed on a single plan."[37] What remains constant is the set of relations and connections between the elements (in this case, the organization of the bones), even as the higher-order effects of these relations (e.g. actual form or function) vary wildly:

> The same formal relations or connections are then effectuated in entirely different forms and arrangements. It is still the same abstract Animal that is realized […], only to varying degrees, in varying modes.[38]

Geoffroy is frequently hailed as an important precursor to the theory of evolution, but Darwinism actually marks something of a divergence from Geoffroy's theory,

Footnote 33 continued
and Deleuze, the only one examined in his book where he takes Deleuze to have a decisive advantage over Hegel.

[34] In fact, the terminological history is slightly more complex than is represented by Deleuze and Guattari. In Geoffroy's writings, "homology" and "analogy" are not carefully distinguished; Richard Owen was the first to strictly separate the two terms, although he does credit Geoffroy (and others, including Goethe, Carus and Oken) with having originally formulated the idea. Owen's terminology is the one widely used both in Deleuze and Guattari's time and today, so they are the terms I will be using here. For further details, see Panchen (1994, pp. 42–44); also Hall (1994, pp. 3–5).

[35] Appel (1987, p. 71), citing Richard Owen's classic and widely-cited formulation.

[37] Appel (1987, p. 69).

[38] Deleuze and Guattari (1987/1980, pp. 45–46/61). See also: 'the organic […] does have a specific unity of composition, a single abstract Animal, a single machine […] and presents everywhere the same molecular materials, the same elements or anatomical components of organs, the same formal connections' (Deleuze and Guattari 1987/1980, p. 45/61).

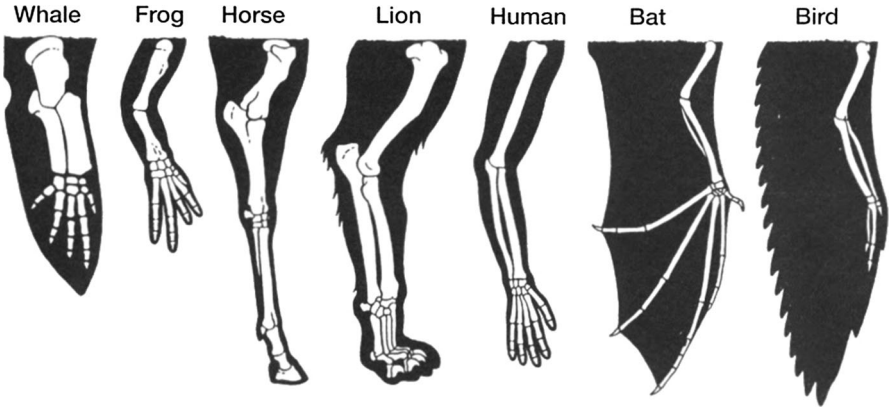


Fig. 1 Diagram of the homologous bone structure of seven vertebrates[36][1]

which is in some ways more interesting. Whereas an explanation of similarities according to the concept of a "common ancestor" would explain the composition of any given animal on the basis of a *similarity* between itself and the foundation or origin out of which it arose, for Geoffroy there is *no resemblance* between this "abstract Animal" (which is not an actual animal, but a virtual plan or schema which describes the order and connection of its bones) and the concrete animals that it effectuates.

To return once again to the terms of Deleuze's ontology, Geoffroy's account—unlike Darwin's—follows its central principle according to which "the foundation can never resemble what it founds [*Jamais le fundament ne peut ressembler à ce qu'il fond*]."[39] This principle emerges out of Deleuze's theory of Ideas which holds, against Plato, that there is no relationship of resemblance between the Idea and the thing of which it is the idea. To return to the example of music, we see an example of a relationship between a foundation—the musical score—and the thing that is founded—the actual performance—whereby there is no resemblance between them. Although the two are in the relationship of foundation-founded, they are nonetheless utterly unlike one another: one is composed of notes written on a page, and the other is the acoustic performance of a piece of music. Returning to Geoffroy's account, we can see that the same principle holds: in Fig. 1, it is not the case that one of these iterations is an "original" in relation to which the others are a derivative "copy," nor is it the case that there is something like a perfect Form of which they are all imperfect instantiations. There is no hierarchy of forms here, or even a genealogical "line of descent." Rather, the Idea of which they are all instantiations is a "map" of the connections, the particular arrangement of parts that they each express in a different way.

[39] Deleuze (1990/1969, p. 99/120).

Geoffroy allows us to see, then, how it is that the organs "organized" by the organism have the capacity to develop in different directions, to allow for the emergence of new actual forms and functions. However, this is only one side of Deleuze and Guattari's theory of the organism. Whilst they do disagree with the traditional conception according to which an organism is a "closed unity," advocating instead an "open" model, the vast majority of their remarks on the organism are *critical* of the concept.[40] An organism is primarily a regularized, organized, hierarchized kind of body, whose organs are restricted to carrying out certain functions prescribed in advance by the organism. As we have just seen, it is *possible* for the organs to do something different, to undergo the kinds of changes in function described by Geoffroy, but this will always go *against* the regularizing forces of the organism as a whole.[41] Bodies will always have the capacity to enter into new combinations to create something new (an "event"), but understanding the body *only* as an organism, without considering everything else it can do, neutralizes those capacities in advance.[42] It is in opposition to the organism that Deleuze and Guattari pit their notorious concept of the "body without organs," to which I now turn.

3 Conclusion: the body without organs

As scholars have noted, the body without organs (sometimes abbreviated to BwO) is a somewhat confusing term, because it does not describe "a body deprived of organs," as the term seems to indicate, but rather "an assemblage of organs freed from the supposedly 'natural' or 'instinctual' organization that makes it an organism."[43] As Deleuze and Guattari put it, for the body without organs, the "enemy" is not the organs, but the organ*ism*, the particular arrangement and configuration of the organs.[44] The body without organs is supposed to designate all

[36] This diagram also makes it clear why Deleuze and Guattari say that Geoffroy "thinks topologically" Deleuze and Guattari (1987/1980, p. 47/63): despite the size and shape of the bones being different in each case, one may move from one to the other by *topological* operations, that is, folding and stretching, but not cutting or joining. These kinds of operations allow one to change the size and shape of the object without changing its virtual form

[40] Somers-Hall (2012, p. 235). It is somewhat surprising that, in this otherwise excellent and extremely illuminating chapter comparing Deleuze and Hegel's theories of the organism, Somers-Hall doesn't once mention that Deleuze's aim is ultimately not to valorize, but to "blow apart" the organism (Deleuze and Guattari 1987/1980, p. 30/43).

[41] As is well known, the process of evolution relies on naturally-occurring mutations, and yet, as Keller interestingly observes: "the vast majority of naturally occurring mistakes are either harmful or neutral" (Keller 2000, p. 32). Mutations, one of the most important sources of change within organisms, are generally not useful, but *harm* the organism, by a massive margin: Keller reports that in bacteria, the estimated ratio of harmful to beneficial mutations is 100,000:1 (Keller 2000, p. 153n. 30). Of course, standard evolutionary theory says that the beneficial mutations are "selected for," but Keller's point is that this only makes sense when one is thinking at the level of the species as a whole. From the perspective of any individual organism, change is *overwhelmingly* harmful.

[42] On this point, see Young et al. (2013, p. 51).

[43] Holland (2013, p. 94).

[44] Deleuze and Guattari (1987/1980, p. 158/196).

of those things that an organic body *could* do, but that it is prevented from doing because of its homeostatic self-regulation processes. The body without organs is the full set of capacities or potentialities of a body prior to its being given the structure of an organism, which only limits and constrains what it can do: it is "what remains when you take everything away."[45] As they ask in *A Thousand Plateaus:*

> Is it really so sad and dangerous to be fed up with seeing with your eyes, breathing with your lungs, swallowing with your mouth, talking with your tongue, thinking with your brain, having an anus and larynx, head and legs? Why not walk on your head, sing with your sinuses, see through your skin, breathe with your belly?[46]

The injunction here is to use our bodies and our organs in ways which are not in thrall to the overarching plan of the organism, to put them to work doing things other than those for which they were designed. In short, to treat them as *machines* capable of producing "events."

The organism, then, can be defined as being a certain way in which the body without organs is "captured," one which restricts its capacities, and constrains it: "the BwO howls: 'They've made me an organism! They've wrongfully folded me! They've stolen my body!'"[47] Of course, "organisms" are not the only way in which the body without organs can be "captured," and in *A Thousand Plateaus*, Deleuze and Guattari analyse a number of other "strata" which impose their own forms on it and limit its capacities.[48] The fact that there are other "strata" helps to explain their otherwise puzzling comment that the body without organs is not an ideal, unattainable point, but something we are attaining all the time.[49] One example is the human face, the subject of an entire chapter of *A Thousand Plateaus*. It is clear that the face is not wholly subordinated to organic functions: we use it to express our emotions, we treat it as an aesthetic object, we use it for communication, and so on. In fact, if one believes the early Levinas, the human face opens us to the very possibility of ethics.[50] All of these functions have nothing to do with the head *qua* organism, and would not have been made possible had the face not first been "freed" from its relation with the organic body and its place within this hierarchy of its system. It is in this sense that the face "removes the head from the stratum of the organism," and thereby frees it to be used in different ways.[51]

Thus, rather than following the conservative tendencies of the organism that always pull it back towards the statistically normal, relegating everything that falls beyond this range to the register of the "pathological," Deleuze and Guattari

[45] Deleuze and Guattari (1987/1980, p. 151/188).

[46] Deleuze and Guattari (1987/1980, pp. 150–151/187).

[47] Deleuze and Guattari (1987/1980, p. 159/197).

[48] The other two "great strata" being significance and subjectification (see Deleuze and Guattari 1987/1980, pp. 159/197).

[49] "What does it mean to disarticulate, to cease to be an organism? How can we convey how easy it is, and the extent to which we do it every day?" (Deleuze and Guattari 1987/1980, pp. 159–160/198).

[50] I thank an anonymous reviewer for pointing out this connection.

[51] Deleuze and Guattari (1987/1980, p. 172/211).

recommend a kind of experimentation whose ultimate goal is the event, that is, the production of something new. And as we saw in the first section, the production of an event changes even the thing that produced the event in the first place, so that the organic body will not remain the same after it has made itself into a body without organs. We humans are able to carry out this kind of experimentation because, as Canguilhem notes, we are fortunate enough to have a surfeit of organs: "too many kidneys, too many lungs, too much parathyroid, too much pancreas, even too much brain, if human life were limited to the vegetative life."[52] Pathological states thus arise not when we use our bodies in ways that make us deviate from the statistical norm, or when we make use of our organs in ways which take them beyond the range of possibilities considered "normal" by the organism, but only when our "experimentation" goes too far, reaching the point where, instead of *increasing* our capacities, it reduces them, and prevents us from creating something new.[53]

Now that we have explored Deleuze and Guattari's modifications to the concepts of "machine" and "organism," let us briefly summarize our findings. Whereas we usually think that machines are defined by their substance, that is, the way in which they are constructed, the form which they take, Deleuze and Guattari understand them according to what they *do*. As they write, a machine should be understood "by *function*, not by form" (recall the example of the knife-rest: understanding it *as a machine* means understanding what it is used for, not its geometric properties).[54] Whereas we usually think that organisms are defined by what they *do*, that is, by their behaviors, by the kinds of activities they carry out, Deleuze and Guattari instead understand them according to their *structure*. As we saw in the distinction between "analogy" and "homology," what makes organisms similar to one another has nothing to do with their function. Rather, comparisons should be based on morphology, that is to say the virtual schema out of which the body emerged. In other words, an organism should be understood by *form*, not by function (recall the example of the bat wing: understanding it *as an organism* means understanding the order and connection of it bones, not what it is used for).[55] Further, in both cases, there is *no substantial link* between the form it has and the function it carries out:

[52] Canguilhem (1991, p. 200).

[53] It is on this point where, despite the great differences between Canguilhem and Deleuze and Guattari, their major concerns nevertheless re-converge. Although he is working with very different conceptions of "machines" and "organisms," Canguilhem is just as implacably opposed to the statistical conception of normality, and for the same reason: concern over the ethical and political consequences of "pathologizing" deviations from the norm. The conception of health developed in *The Normal and the Pathological* combats this "normalizing" view using the very same strategy as Deleuze and Guattari do: rejecting statistical norms as a criteria, Canguilhem instead focuses on the production of the new, defining health as the capacity to create *new* norms in *new* situations.

[54] Deleuze and Guattari (1987/1980, p. 141/176).

[55] Although it should be noted that by "form" I here mean *virtual* form—that is, the "formal relations or connections" which connect the various concrete forms that existing organisms may take – and not the *actual* form taken by each individual organism.

[56] The reverse is also the case: one would not arrive at the precise structure of a bat wing from the mere idea of an "organ whose purpose is flying," nor would one arrive at the geometrical properties of the knife-rest from the mere idea of "a machine whose purpose is to hold knives." In both cases, there is a multiplicity of other ways that they could have been structured which would still have allowed them to carry out this same purpose. The reason why they have one structure rather than another comes not from

one can no more deduce the function of a bat's wing from its morphology than one can deduce the function of a knife-rest from its geometrical properties.[56]

There is something like a priority of creation in Deleuze and Guattari, a preference for the new, which leads them away from what might otherwise appear to be a kind of symmetry between the two concepts (function not form vs. form not function), towards a valorization of the idea of the machine, and a strong criticism of the idea of the organism. This leads, first, to an asymmetry between the scope of the two concepts: whereas their idea of "machine" is supposed to be universal (*everything* is a machine), their idea of "organism" is restricted to a certain kind of body. But perhaps more importantly, it also leads them to a different understanding of the *relation* between the two terms, centered on their concept of the "body without organs." Their non-mechanical mechanism, which is also a vitalism of the inorganic, highlights not the form or structure that bodies *actually have*, but rather the virtual capacities that bodies have to do something different. A body may be structured like an organism, but, since its organs are all machines, it will always retain the capacity to "disarticulate," as they put it, to cease to be an organism. The body without organs, then, can be defined as the *becoming-machine of the organism*; it is what happens when one part of the body enters into combination with some other machine in a way which allows it to escape from the organism's regularizing, normalizing processes. Seen in this way, the body previously considered an organism is opened up to a whole host of new connections, each of which may lead to the production of an event.

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Footnote 56 continued
their "purpose," but from the specific process of actualization that produced them (the specifics of evolutionary history in the case of the bat; the personal taste of the designer in the case of the knife-rest).

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